Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 9 Tutorial Worksheet 1: VPN Foundations and Types

Field	Detail
Based on	Week 9 Deck A, VPN Foundations lecture (nature of the internet, VPN definition, tunnel vs VPN, software vs hardware, the five VPN types)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate and compare security technologies)
Format	12 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's VPN Foundations lecture. Several ask you to recommend or compare rather than simply describe, which is deliberate: LO4 is specifically about evaluating and comparing security technologies, not just naming them. Write full answers, not one-line definitions.

Foundations

Q1. Explain why the internet's fundamental design creates a need for VPNs, rather than confidentiality being an inherent property of the network itself. *[LO3]*

Q2. Explain what a VPN is, and describe the three things it combines to achieve confidentiality, integrity, and authentication. *[LO3]*

Q3. Explain the difference between a tunnel and a VPN. Give a specific real-world example of a tunnel that is not a VPN, and explain why it fails to qualify as one. *[LO3, LO4]*

Q4. GRE over IPsec combines a tunnel with a VPN technology. Explain what each component contributes to this combination, and why using GRE alone would not be sufficient for a security-conscious organisation. *[LO4]*

Classification

Q5. Explain the difference between the deployment form axis (software versus hardware) and the connection purpose axis used to classify VPNs this week. Why are these treated as independent questions rather than a single list of types? *[LO3, LO4]*

Q6. A company is deciding between a software-based and a hardware-based VPN gateway for a site connecting hundreds of employees' traffic. Evaluate which deployment form is more appropriate, and justify your answer using at least two dimensions such as cost, performance, or scalability. *[LO3, LO4]*

The Five VPN Types

Q7. Explain why Personal VPN and Mobile VPN are both described as variants of Remote Access VPN, rather than being entirely separate categories. *[LO4]*

Q8. Explain why Cloud VPN is described as a variant of Site-to-Site VPN specifically, rather than Remote Access. *[LO4]*

Q9. A logistics company has drivers using tablets that must remain connected to a central dispatch system while moving between cellular and WiFi networks throughout the day. Recommend the most appropriate VPN type for this scenario, and justify your recommendation using at least two specific requirements from the scenario. *[LO3, LO4]*

Q10. A financial services firm needs to connect its headquarters to three branch offices with a permanent, always-on connection carrying high transaction volumes, with no individual staff member managing their own VPN connection. Recommend the most appropriate VPN type, and explain what would go wrong if Remote Access VPN were used instead. *[LO3, LO4]*

Q11. Compare Personal VPN and Remote Access VPN. Both involve an individual user connecting through a tunnel, so what is the key difference in what each one is actually connecting to, and why does that difference change where the user's trust has to be placed? *[LO4]*

Q12. A manufacturing company wants to connect its on-premises factory network to a data analytics platform hosted by a cloud provider, with continuous automatic data flow and no physical VPN hardware to manage inside the cloud provider's own infrastructure. Identify the most appropriate VPN type and justify your answer against the other Site-to-Site option. *[LO3, LO4]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. Several questions above ask you to recommend one VPN type over another for a specific scenario. Practise stating your recommendation first, then justifying it against the scenario's actual requirements, since that is exactly the structure a structured-essay exam question rewards.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).